TeleComX Telecom — Customer Churn Analysis

Executive Summary

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Tools Used: Python · SQL · Power BI

Dataset: 7,043 customer records

Business Problem

TeleComX is experiencing accelerating customer churn, rising from 18% to 26.54%. Every churned customer represents lost monthly revenue, increased acquisition costs, and reduced customer lifetime value. The Business Intelligence team was tasked with identifying why customers leave, which customers are most at risk, and what the business should do to retain them.

Approach

The analysis followed a structured five-phase methodology:

1. ***Data Understanding*** — Inspected 7,043 records across 21 variables. Identified and resolved a data quality issue in the TotalCharges column where 11 blank entries were stored as text rather than null values.

2. ***SQL Analysis*** — Wrote 8 business queries to answer key management questions about churn drivers across contract type, payment method, internet service, tenure, and customer demographics.

3. ***Exploratory Data Analysis*** — Produced 7 visualizations confirming SQL findings and revealing the relationship between monthly charges, tenure, and churn behavior.

4. ***Predictive Modeling*** — Built and compared three machine learning models. Logistic Regression achieved the strongest performance with 80.7% accuracy and 84.19% ROC-AUC score, correctly identifying 211 at-risk customers in the test set.

5. ***Retention Strategy*** — Translated all findings into 5 prioritized business recommendations with expected impact estimates.

Key Findings

1. Churn is concentrated in a specific customer profile

Month-to-month customers on fiber optic internet paying by electronic check churn at **60.37%** — more than double the company average.

2. Contract type is the single strongest retention lever

Two-year contract customers churn at just **2.83%** vs **42.71%** for month-to-month customers. The difference is dramatic and actionable.

3. New customers are the highest risk group

Customers in their first 12 months churn at **47.44%**. The business is losing nearly half its new customers before they become profitable.

4. Churned customers pay more

Customers who left paid an average of **$74.44/month** vs **$61.27** for customers who stayed. This is a revenue quality problem, not just a volume problem.

5. Protective services dramatically reduce churn

Customers without tech support churn at ~40%. Customers with tech support churn at ~15%. Bundling these services is a low-cost, high-impact retention tool.

Model Performance

Metric	***Result***
Model Selected	Logistic Regression
Accuracy	80.70%
ROC-AUC	84.19%
Customers correctly flagged as at-risk	211 out of 374

The model provides TeleComX with the ability to identify at-risk customers **before** they churn, enabling proactive intervention rather than reactive damage control.

Recommendations

Priority	*Action*	*Expected Impact*
🔴 Immediate	Offer contract upgrade incentives to month-to-month customers	Reduce churn by ~8%
🔴 Immediate	Incentivise switch from electronic check to automatic payments	Prevent 200–300 churns/month
🔴 Immediate	Launch 90-day structured onboarding for new customers	Retain ~300 additional customers/year
🟡 Short-term	Investigate fiber optic service quality and satisfaction	Preserve $1M+ annual revenue
🟢 Ongoing	Bundle tech support into high-risk customer packages	Reduce segment churn by 5–8%

Conclusion

TeleComX's churn problem is serious but solvable. The data shows that churning customers are not randomly distributed — they share identifiable characteristics that make them predictable and therefore preventable. With targeted interventions focused on contract migration, payment method incentives, and new customer onboarding, TeleComX can realistically return churn to below 18% within 12 months, preserving millions in annual recurring revenue.